



Interaction of glucosamine with uracil and thymine: a computational study

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Received: 16 June 2022 / Accepted: 22 August 2022 / Published online: 26 August 2022

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Abstract

The present study delves into the interaction of the monomer of glucosamine with uracil and thymine in vacuum and in different phases by density functional theory (DFT). Both the molecular geometries were optimized at B3LYP with a 6-31G(d,p) basis set. The binding energy, interaction energy, and solubility were calculated from the optimized molecular geometry. The dipole moment and the electronic energies were found of the optimized product in different solvents (water, ethanol, methanol, heptane, cyclohexane, and CCl₄) which describes the solubility of the interactive molecule in polar and non-polar solvents. The electronic energies are nearly the same for all the solvents. Observed theoretical results are expected to guide future relevant experimental research on gene delivery by glucosamine. This will also help in enhancing pharmaceutical research as carrier drug delivery, tissue repair, gene delivery, spermicidal activity, anti-tumor, and anti-microbial resistance.

Keywords Density functional theory (DFT) · Glucosamine · Uracil · Thymine · Hydrogen bonding · Molecular orbitals (MOs) · Solubility

Introduction

Glucosamine declines the levels of enzymes which degrades cartilage and helps in repairing damaged tissues [1]. It is found that in a physiological environment, glucosamine can readily be digested by chitinases which can be produced normally in the human intestine or exist in the blood streams [2]. Single-stranded regions of DNA and RNA play an essential role in numerous processes in living cells. According to previous studies [1–4], thymine derivatives synthesize anti-cancer effects. Therefore, the fusion of glucosamine may broaden its biomedical utility to include anti-cancerous action [3]. As the size of this molecule is small, it can easily be passed through the biological barrier and deliver drugs. Therefore, it has been used in pharmaceutical research as a carrier drug delivery, tissue repair, gene delivery, spermicidal activity, anti-tumor, and anti-microbial resistance [5–8]. Ribonucleic acid (RNA) is a single-stranded molecule. An RNA strand

has a backbone made up of ribose sugar and a phosphate group. To each sugar, there is a nitrogenous base attached to it. The four nitrogenous bases are adenine (A), cytosine (C), uracil (U), and guanine (G). RNA was the first genetic material. All essential life processes like metabolism evolve around RNA. Along with a generic material, RNA is used as a catalyst (important biochemical reactions in the living system are catalyzed by RNA). The single-stranded DNA, as well as RNA, plays a vital role in many processes in living cells. RNA interference has now become a successful and powerful tool for gene expression in living cells [4]. Deoxyribonucleic acid (DNA) is a double helix molecule. It is an important hereditary material present in all living organisms. Approximately every cell in a living body has the same DNA. The information stored in DNA is in the form of code. The codes are made up of nitrogenous bases which are adenine (A), cytosine (C), guanine (G), and thymine (T). DNA bases pair up with each other to form units called base pairs. Adenine with thymine and guanine with cytosine form two base pairs of DNAs. Glucosamine is a linear monosaccharide. It contains an amino group at the place of one of the hydroxyl groups. Like most saccharides, glucosamine has a cyclic configuration. It is obtained by chitin which is also a polysaccharide [5]. Glucosamine is non-toxic in both

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animals and humans [6]. Its sulfate salt is used in osteoarthritis. Osteoarthritis (OA) has some symptoms, but the disease is quite common, and its impact is quite large [7]. Glucosamine seems to have a weak pain-relieving effect. Though it does not reduce the risk of getting osteoarthritis, still it provides some pain relief and improves function.

One of the four nucleobases in the nucleic acid of RNA is the frequently occurring pyrimidine derivative known as uracil. Through two hydrogen bonds in RNA, it binds to adenine. In DNA, thymine takes the place of the uracil nucleobase. Uracil can be thought of typically as a demethylated version of thymine. Modeling and forecasting the behavior of these nucleic acid bases (NABs) of DNA and RNA in bigger compounds, such as nucleosides, nucleotides, and DNA itself, require a thorough understanding of their geometries [8]. Additionally, uracil is regarded as a key structural element in drug discovery due to its extensive range of biological activity and synthetic accessibility. From a biological perspective, systematic experimental and theoretical studies of nucleic acid bases and their derivatives are extremely important. In order to create soluble ribonucleic acid, uracil and its derivatives are required. They are also utilized to create anti-cancer and anti-HIV drugs [9]. The two most frequently mentioned properties of uracil and its derivatives are antiviral and anti-tumor. To learn more about the functionality and applications of these two molecules, numerous theoretical and experimental investigations have been conducted on them [9, 10]. A comparison of the fundamental geometrical and electronic characteristics of these two nucleobases with glucosamine in gas phase and solvents, however, has not been done in a single study. In order to observe how the environment (gas and solvent) affected the molecular and electronic properties of uracil and thymine when they interacted with glucosamine, interaction of uracil and thymine with glucosamine was optimized DFT levels of theory.

The present study is an attempt to study the reactivity of glucosamine with uracil and thymine which is one of the nitrogenous bases of RNA and DNA using density functional theory. DFT is a theory that helps to solve the theory of the nature of electron correlation. It is used by many researchers to know much about many different processes at the molecular and atomic levels. The interesting fact about them is that they both are absent from the other's genetic material. In simple words, uracil is present in RNA but absent in DNA, whereas thymine is present in DNA but absent in RNA. The reactivity of glucosamine with uracil and thymine is further studied in water, methanol, ethanol, heptane, cyclohexane, and CCl_4 . In the previous theoretical studies [11] using DFT, the reactivity of glucosamine and its derivatives with guanine were reported. The electronic structure and geometries of uracil cation and fragmentation

were also reported. To the best of the author's knowledge, these results are not reported yet.

Computational study

Density functional theory is used in this present study. The molecular geometries of the monomer of the deacetylated form of the glucosamine, thymine, and uracil were optimized by the calculation method of Becke's three-parameter exchange and Lee, Young, and Parr correlation functional (B3LYP) with the basis set of 6-31G(d,p) of DFT [12–14]. The molecules were firstly individually optimized under B3LYP with the basis set of 6-31G(d,p) [15]. They were further optimized together (glucosamine with thymine and uracil) with the same calculation method and basis set. The charge on this optimized product was zero with a singlet spin. The binding energies were calculated at the given theory. For solvation effects, different solvents were used in the optimized interacted geometries of glucosamine, thymine, and uracil. The solvents were categorized into polar and non-polar form. The solvents used were water, methanol, ethanol, chloroform, CCl_4 , and heptane. The geometries were optimized at B3LYP/6-31G(d,p) level [11]. For the study of thermochemistry [16], the optimized interacted product was put into the job of the frequency with the same calculation method and basis set. From the frequency calculation, different parameters such as entropy, temperature, and pressure were determined. All the calculations were performed using Gaussian09 with the Gauss View program to visualize the molecular structures [12]. The chemical structures of glucosamine, uracil, and thymine are shown in Fig. 1.

Results and discussions

Optimized structures and interactions

In the present study, the interaction of glucosamine with uracil and thymine is theoretically investigated. The optimized chemical structure of glucosamine with uracil and thymine is shown in Fig. 2.

Figure 2a shows the optimized structure of glucosamine + uracil. For the interaction of both the molecules, particular sites were favored. The H18 site of the glucosamine interacts with the O30 site of uracil via hydrogen bonding. The formation of hydrogen bonding depends on the position and nature of acceptor and donor atoms. The length of the hydrogen bond must be smaller than the total of the van der Waals (vdW) radii of the hydrogen atom and any other atom that forms the bond, depending on the properties of hydrogen bonds. Since the vdW radii of the hydrogen and oxygen atoms are 1.2 Å and 1.52

Fig. 1 Chemical structure of glucosamine, uracil, and thymine

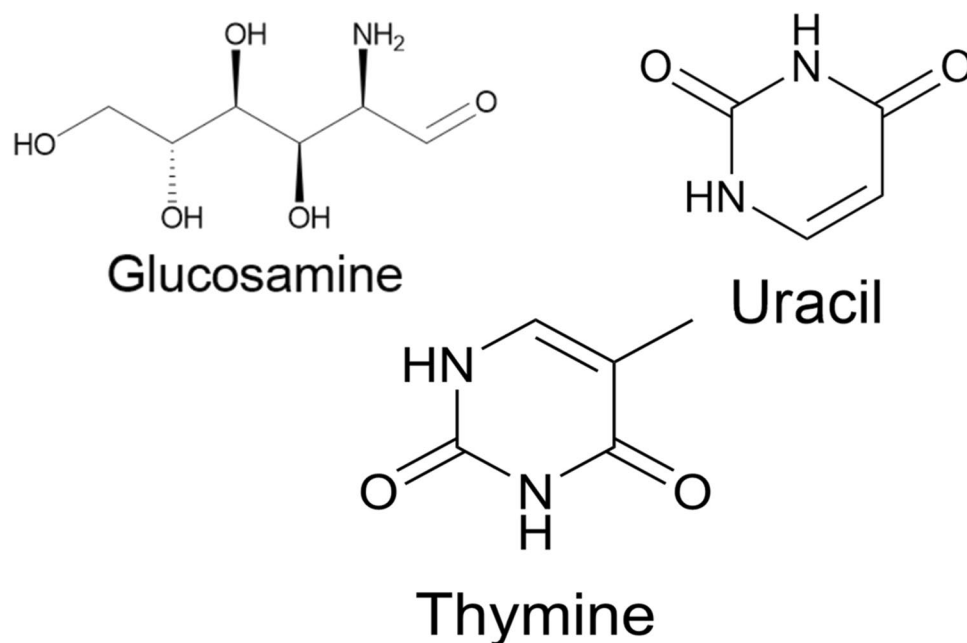
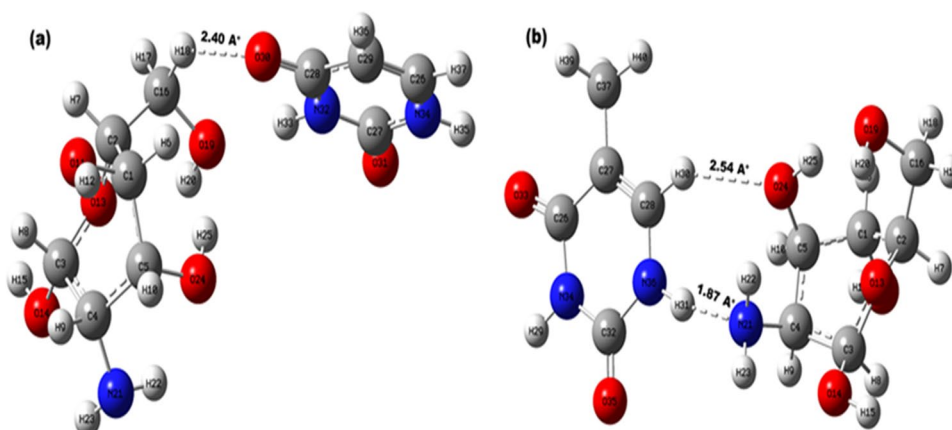


Fig. 2 Optimized structure of **a** glucosamine with uracil and **b** glucosamine with thymine



$\text{\AA}^\circ$, respectively, the length of the H—O bond should be less than $2.72 \text{ \AA}^\circ$ [17, 18]. The bond length between H18 and O30 is obtained to be $2.402 \text{ \AA}^\circ$ which is less than $2.72 \text{ \AA}^\circ$ which confirms the presence of a hydrogen bond. Such type of bond can generally occur in both organic and inorganic molecules. Hydrogen bonding is stronger than van der Waals interaction but weaker than covalent bonds. The presence of hydrogen bonding shows the strong interaction between glucosamine and uracil. This special hydrogen bonding plays a very important role in the binding and release of nucleic acid during gene delivery [13]. The electronic energy and dipole moment of the optimized molecule is noted to be $-679,069.343 \text{ kCal/mol}$ and 5.9681 Debye , respectively. For the determination of binding energy, optimized monomers are required. Hence, the binding energy (ΔE) is calculated using Eq. 1 [14]:

$$\Delta E = E_{(G)} + E_{(U)} - E_{(G+U)} \quad (1)$$

where $E_{(G)}$ = electronic energy of glucosamine.

$E_{(U)}$ = electronic energy of uracil.

$E_{(G+U)}$ = electronic energy of optimized product of glucosamine and uracil.

The binding energy of the product was calculated to be 18.198 kCal/mol . The binding energy was found to be a positive value which means the interaction is endothermic. The interaction energy of the optimized product was calculated using Eq. 2 [28]:

$$E_{\text{interaction}} = E_{(G+U)} - E_G - E_U \quad (2)$$

and was calculated to be -18.198 kCal/mol . From the thermochemical study, it was found that at room temperature

(298.150 Kelvin) and pressure (1.00 atm) the molecules interact together. The polarizability(α) was calculated to be 148.078000 a.u [19].

Furthermore, the interaction of glucosamine with thymine was theoretically investigated using density functional theory. Both the molecules were initially optimized individually, and then, they interacted together. Sites for the interaction were favored. After the interaction, it was seen that the final structure was distorted which depicts the change in bond length and bond angle. Figure 2b shows the interaction between glucosamine and thymine. From Fig. 2b, it can be seen that between H40 and O24, hydrogen bonding is present. The bond length between O24 and H40 was obtained as 2.54 Å which is less than 2.72 Å confirming the presence of a hydrogen bond. Similarly, the hydrogen bond of length 1.87 Å is observed between H31 and N21 of the glucosamine-thymine complex. The binding of glucosamine to nucleobases has been shown to be caused via a secondary interaction, such as hydrogen bonding. Hydrogen bonding plays a crucial part in the formation of adducts of glucosamine and nucleobases has also been as previously reported by Deka et.al [13]. Herein, theoretical evidence of hydrogen bonding between glucosamine and nucleobases focusing on uracil and thymine as a representative case is pursued.

The electronic energy and dipole moment of the glucosamine-thymine are noted to be $-703,740.7465486$ kCal/mol 10.009 Debye, respectively. For the calculation of binding energy, the optimized molecules were used. Hence, the binding energy (ΔE) is calculated using Eq. 3:

$$\Delta E = E_{(G)} + E_{(T)} + E_{(G+T)} \quad (3)$$

where $E_{(G)}$ = electronic energy of glucosamine.

$E_{(T)}$ = electronic energy of thymine.

$E_{(G+T)}$ = electronic energy of optimized product of glucosamine and thymine.

The binding energy of the product was calculated to be 13.934 kCal/mol. The binding energy was found to be a positive value which means the interaction is endothermic. The interaction energy of the optimized product was calculated using Eq. 4:

$$E_{interaction} = E_{(G)} + E_{(T)} - E_{(G+T)} \quad (4)$$

and was calculated to be -13.934 kCal/mol. From the thermochemical study, it was found that the interaction between both the molecules takes place at room temperature (298.150 Kelvin) and pressure (1.00 atm). The polarizability (α) was calculated to be 158.083667 a.u.

HUMO-LUMO study

One of the most popular reactivity descriptors used by researchers to assess the chemical stability of a system is the

highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO) [20]. As such, HOMO–LUMO distribution within the considered molecular complexes is evaluated to study their chemical stability. As indicated in Fig. 3, the HOMO and LUMO orbitals are distributed over glucosamine showing its strong interaction with the nucleobases. The study of molecular orbital describes the occupied and unoccupied molecular orbital of the optimized structure. The HOMO and LUMO of the interactive molecules discuss the strength and stability of the orbitals. The HUMO-LUMO gap is a proportion of electron jumps from occupied and unoccupied orbitals [20]. The HUMO and LUMO state of glucosamine can be shown in Fig. 3a. The energy E_{HOMO} of glucosamine is -5.868 eV, whereas the energy E_{LUMO} of glucosamine is 1.429 eV. The energy gap between the HOMO and LUMO can be determined by the following Eq. 5 [25, 26]:

$$E_g = E_{LUMO} - E_{HOMO} \quad (5)$$

From the above equation, the energy gap of glucosamine was calculated to be 7.297 eV.

The energy gap of glucosamine with uracil and thymine was determined using the above Eq. 5. The HOMO–LUMO state of glucosamine with uracil is shown in Fig. 3b, whereas the state of glucosamine with thymine is shown in Fig. 3c. The energy gap (E_g) of glucosamine-uracil and glucosamine-thymine is determined to be 5.673 eV and 5.465 eV, respectively. It is noted that the energy gap of glucosamine is greater than the energy gaps when interacted with RNA and DNA bases. It is also noted that when glucosamine interacts with RNA and DNA bases, then the energy difference between them is quite less. The lower the energy gap, the more stable the compound might become.

Solvation study

The characteristics of a molecular system in the presence of solvents can be computed using two different computational procedures. In the first, all systems are optimized in the gas phase, and then solvents are added thereafter during single-point calculations. The second one entails re-optimizing the systems while solvents are present. The second procedure is followed in this; i.e., the geometrical minima of glucosamine-uracil and glucosamine-thymine are further ascertained in ethanol, methanol, heptane, cyclohexane, and CCl_4 and in water media. The optimized chemical structure was studied in different solvents having varied dielectric constant such as water ($\epsilon = 78.36$), methanol ($\epsilon = 32.70$), ethanol ($\epsilon = 25.30$), cyclohexane ($\epsilon = 2.02$), heptane ($\epsilon = 1.92$), and CCl_4 ($\epsilon = 1.74$). The energies were calculated in these solvents using the PCM model with the geometries optimized at B3LYP/6-31G(d,p) level [11]. Solvents affect the physical

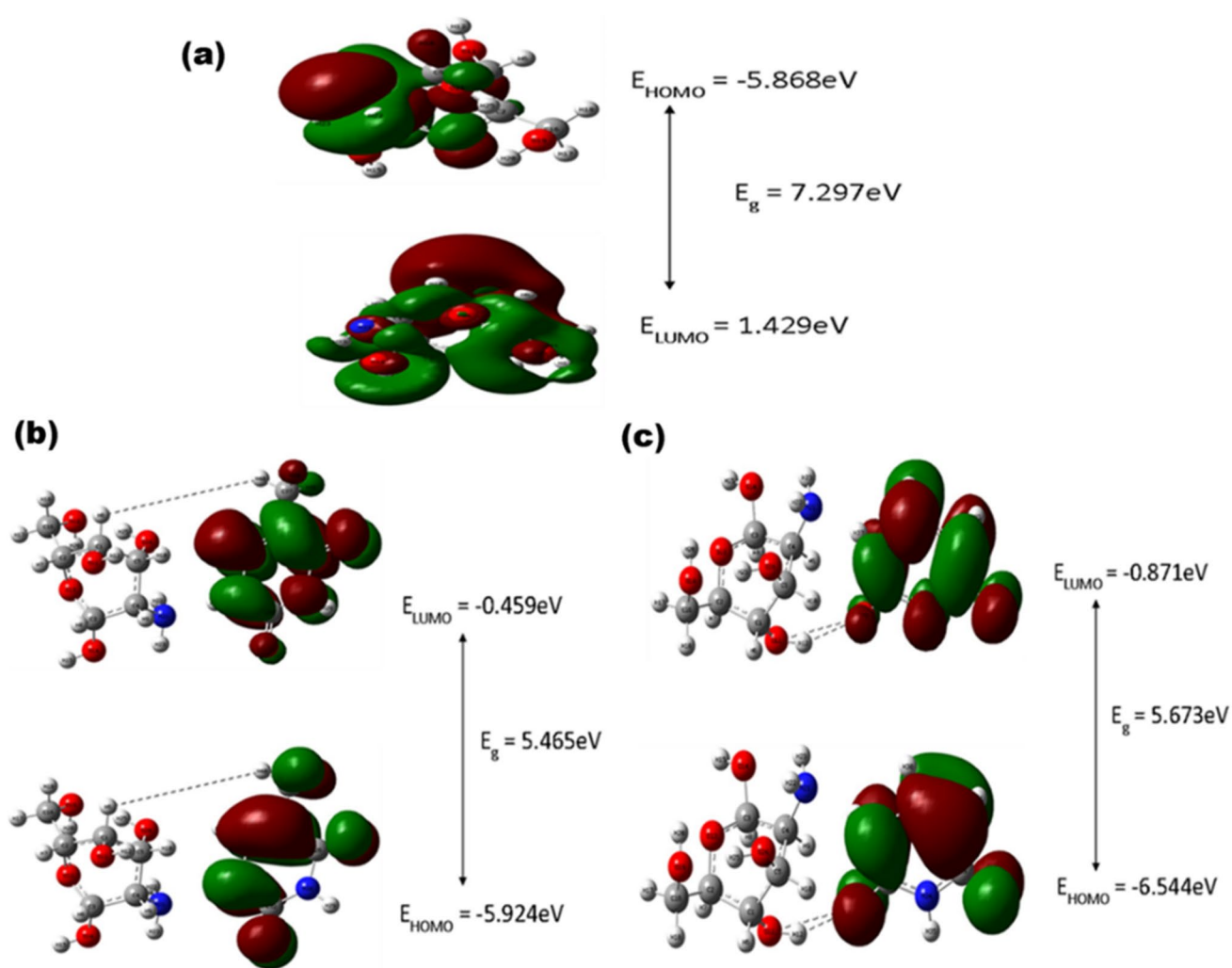


Fig. 3 HOMO–LUMO gap of **a** glucosamine, **b** interaction of glucosamine with uracil, and **c** interaction of glucosamine with thymine

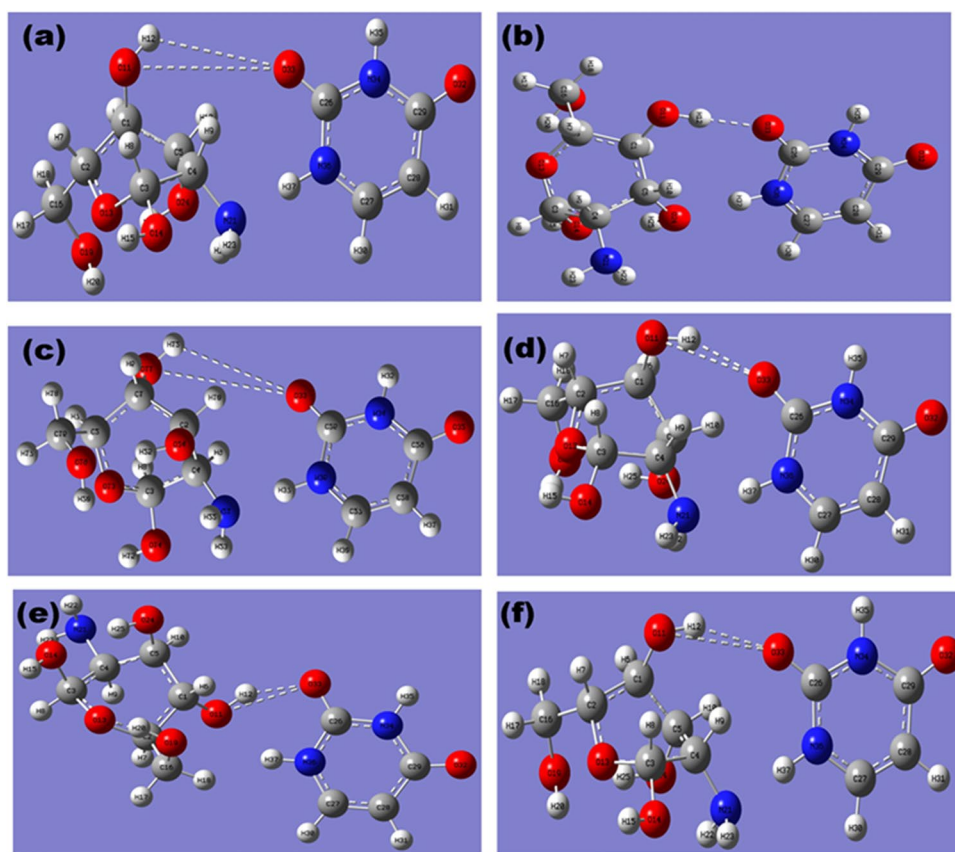
and chemical properties and reactivity of any organic molecule [17]. The solute–solvent interactions are smaller as compared to the intramolecular forces. On account of this, it is expected that the solute's geometry does not undergo important changes. Hence, geometry optimization is obligatory when properties are computed in the solution.

The optimized structure of glucosamine–uracil and glucosamine–thymine in various polar and non-polar solvents are shown in Fig. 4 and Fig. 5, respectively.

The optimized structure shows variation in interaction in polar and non-polar solvents. Though the bonding seems to be the same in the solvents, they have different energies and dipole moments as reported in Tables 1, 2, 3, and 4. To facilitate polar solvents, there are some geometrical parameters such as bond angle whose deviation can cause considerable changes of permanent electric moments which can appear on these parameters. These properties of chemical processes in solutions are usually detailed employing free energy

variations. Figure 6 shows the variation between electronic energy and dipole moment of glucosamine + uracil in a polar and non-polar solvent. Figure 6a and 6b represents the solubility and stability of the optimized product in different solvents. The variation between electronic energy and dipole moment of glucosamine + thymine in the polar and non-polar solvent is shown in Fig. 7. From all the graphs it is noted that in each interaction, in water the interaction shows the highest dipole moment with high electronic energy. This proportionality clears the fact that the interacted molecules are particularly soluble in water. It shows that electronic energy plays a vital role in the solubility of molecules. For non-polar solvent, it can be observed that the dipole moment of CCl_4 is high, but their electronic energy is low in both the interactions. Since the solubility of the molecule is dependent on its high electronic energy, hence it can be said that the interacted molecules are least soluble in CCl_4 .

Fig. 4 Optimized structure of glucosamine + uracil in various solvents: **a** water, **b** CCl_4 , **c** methanol, **d** cyclohexane, **e** ethanol, **f** heptane



Comparison between electronic energy of complexes in polar and non-polar solvent

This comparative study discusses the electronic energy of both glucosamine-uracil and glucosamine-thymine in polar (Table 5) and non-polar solvents (Table 6). As discussed previously, electronic energy plays an efficient role in explaining the solubility of any interacted molecule. This column is a comparative study between both the nitrogenous bases in polar and non-polar solvents which further explain their solubility in the given media. The structure of RNA is unstable, while the structure of DNA is stable. Due to this fact, RNA changes its structure when combined with any other media. This explains clearly that when the nitrogenous base of RNA (uracil) shows interaction with glucosamine and then further combined with solvents, it changes its structure and becomes stable. The change in the structure becomes more rapid when a polar media (water) is used rather than a non-polar media. Since DNA is stable, therefore its base does not change its structure rapidly. But when interacting with another molecule, it also shows the change in its nature. Hence, it also gets soluble when solvents are used. From Fig. 8a and 8b, the electronic energy is more negative in glucosamine-thymine rather than glucosamine-uracil in both

polar and non-polar solvents. Subsequently, the electronic energy of glucosamine-uracil is higher than glucosamine-thymine which clears the fact that uracil has a higher solubility than thymine in polar as well as a non-polar solvent. In addition to this, the solubility of both glucosamine-uracil and glucosamine-thymine is higher in polar solvents than in non-polar solvents.

Comparison between dipole moment of complexes in polar and non-polar solvent

This comparative study shows the variation in dipole moments of glucosamine-thymine and glucosamine-uracil in different solvents. The value of the dipole moment is shown in Table 7 and Table 8. Dipole moment is the calculation of charge separation in any molecule. In simple words, it can be depicted that the greater the difference in electronegativities of bonding atoms, the greater will be the dipole moment of the molecule [18]. A previous study shows that the dipole moment of a huge water cluster can be traced only when there is a considerable electron delocalization along with hydrogen bonds [18]. Theoretical studies disclose the increase of dipole moment of a water molecule. From the data given below, the interaction

Fig. 5 Optimized structure of glucosamine + thymine in various solvents: **a** water, **b** CCl₄, **c** methanol, **d** cyclohexane, **e** ethanol, **f** heptane

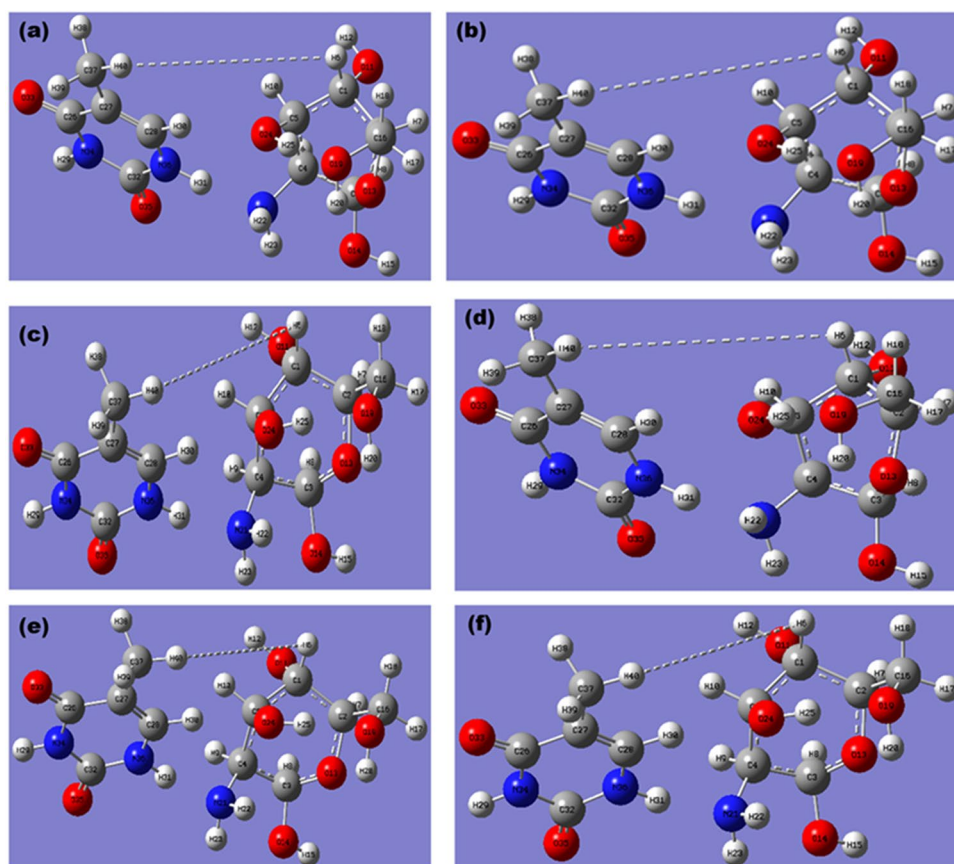


Table 1 Interaction of glucosamine with uracil in polar solvents

Solvents	Dielectric constant	Electronic energy (Kcal/mol)	Dipole moment (Debye)
Water	78.36	−679,014.010	11.45
Methanol	32.70	−679,081.817	9.85
Ethanol	25.30	−679,078.677	9.21

Table 2 Interaction of glucosamine with uracil in non-polar solvents

Solvents	Dielectric constant	Electronic energy (Kcal/mol)	Dipole moment (Debye)
CCl ₄	2.24	−679,075.555	8.30
Cyclohexane	2.02	−679,074.105	6.78
Heptane	1.92	−679,073.790	6.73

Table 3 Interaction of glucosamine with thymine in polar solvents

Solvents	Dielectric constant	Electronic energy (Kcal/mol)	Dipole moment (Debye)
Water	78.36	−703,756.731	12.55
Methanol	32.70	−703,756.292	12.48
Ethanol	25.30	−703,755.539	11.28

Table 4 Interaction of glucosamine with thymine in non-polar solvents

Solvents	Dielectric constant	Electronic energy (Kcal/mol)	Dipole moment (Debye)
CCl ₄	2.24	−703,747.946	11.16
Cyclohexane	2.02	−703,747.256	11.06
Heptane	1.92	−703,746.817	10.99

of molecules in water is highest rather than in any other solvent. To calculate the dipole moment, the electronic density with respect to the orbital of the molecule needs to be investigated. It has been noted that these orbitals are

not uniquely determined and are dependent on localization criteria. The graph in Fig. 9a and 9b shows that the dipole moments of both the nucleic bases vary from each other. It is seen that the dipole moment of thymine is greater than

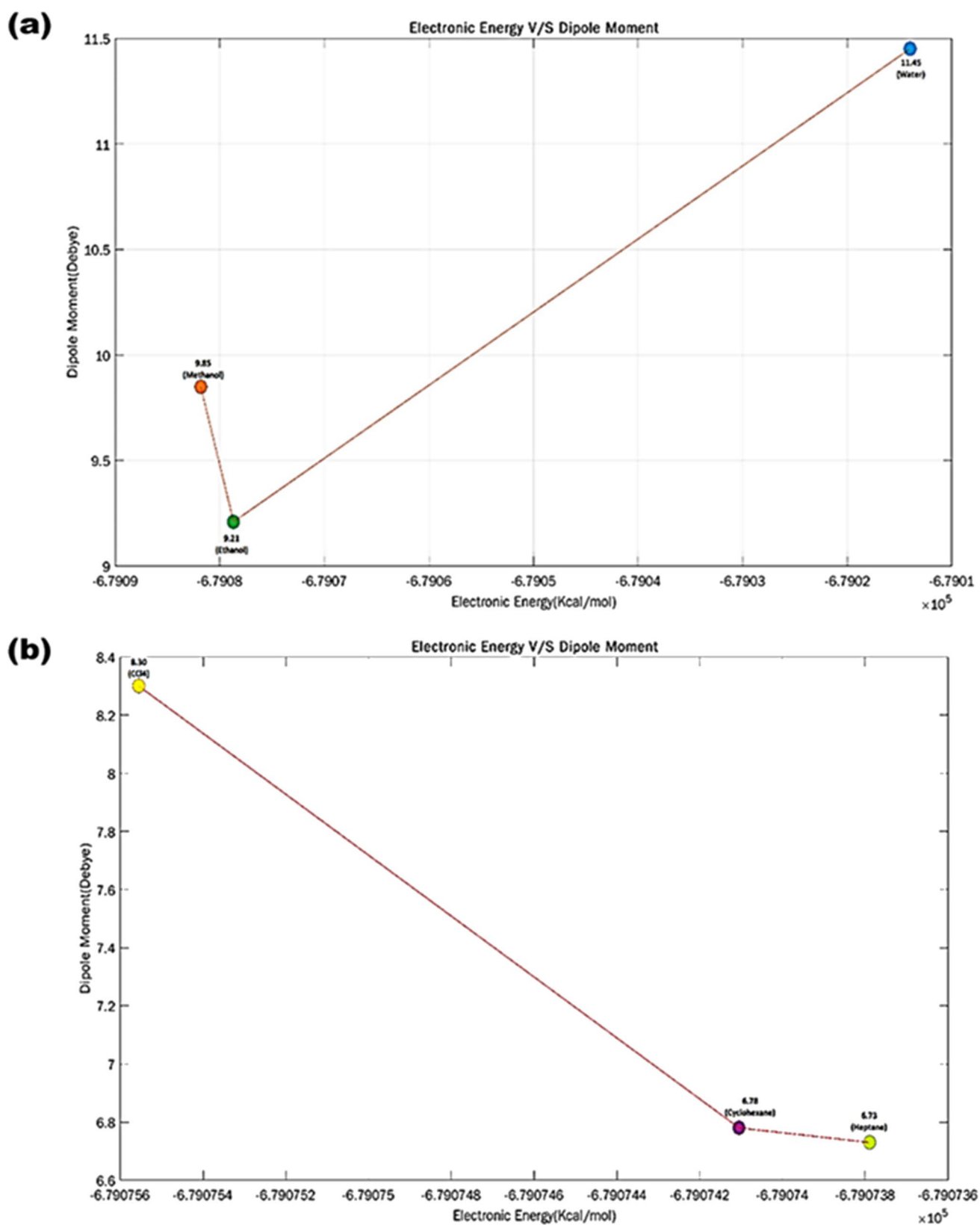


Fig. 6 Variation between electronic energy and dipole moment of glucosamine + uracil in **a** polar solvents and **b** non-polar solvents

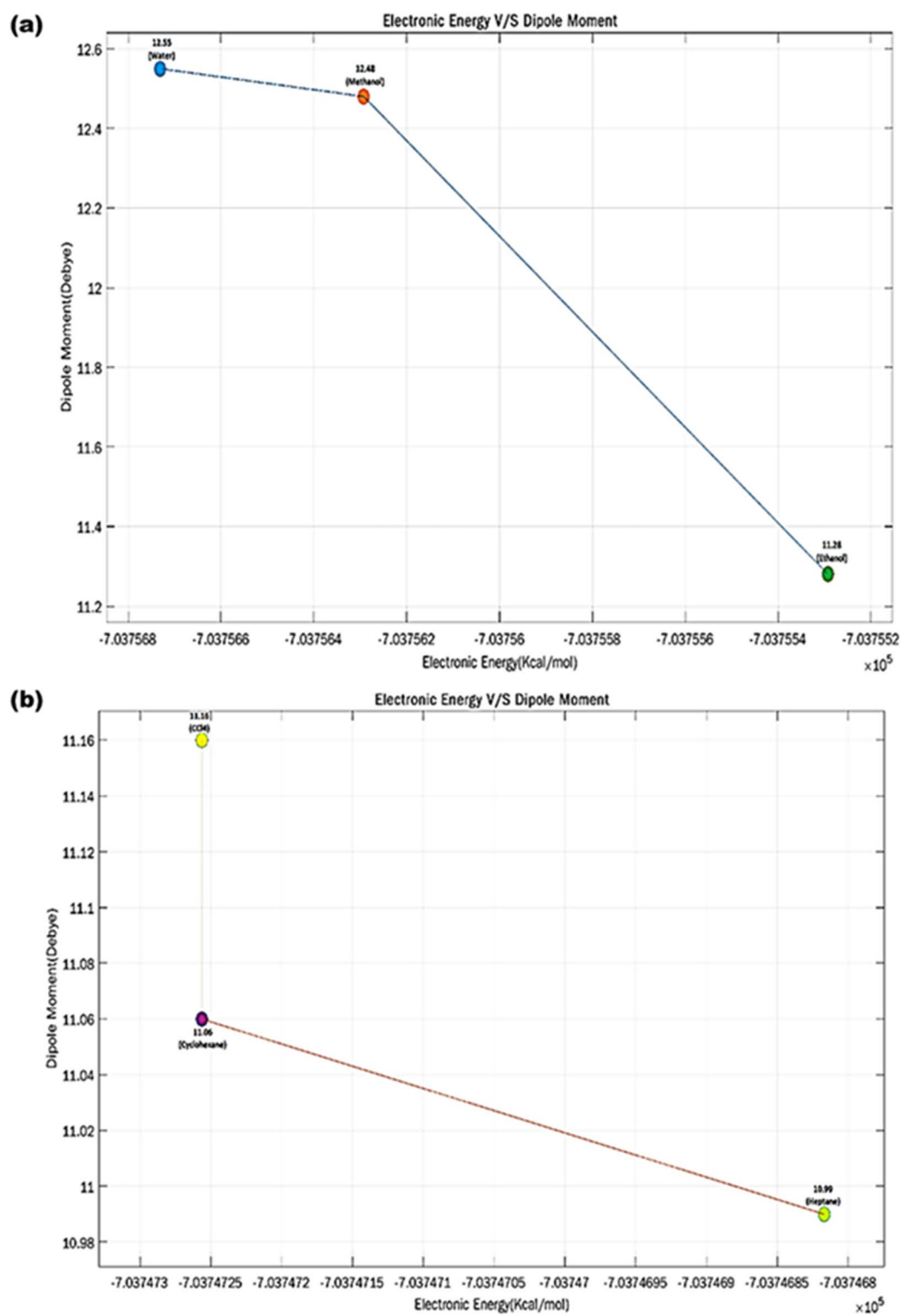


Fig. 7 Variation between electronic energy and dipole moment of glucosamine+thymine in **a** polar solvents and **b** non-polar solvents

Table 5 Electronic energy in polar solvents

Solvents	Electronic energy (Glucosamine-thymine)	Electronic energy (Glucosamine-uracil)
Water	−703,756.731	−679,014.010
Methanol	−703,756.292	−679,081.817
Ethanol	−703,755.539	−679,078.677

Table 6 Electronic energy in non-polar solvents

Solvents	Electronic energy (Glucosamine-thymine)	Electronic energy (Glucosamine-uracil)
CCl ₄	−703,747.946	−679,075.555
Cyclohexane	−703,747.256	−679,074.105
Heptane	−703,746.817	−679,073.790

uracil in both polar and non-polar solvents. A steady drop in dipole moment is observed with the decrease in polarity of the medium.

Conclusion

The prime object of this study is to provide an insight into the reactivity and interaction of glucosamine with nucleobases. Two nucleobases uracil and thymine were chosen for this study. Our study confirms the presence of hydrogen bonding between glucosamine and nitrogenous bases (uracil and thymine as a representative case in this study). The interaction energy of glucosamine with both the nitrogenous bases is endothermic. Electronic energy and solubility of the complexes both are sensitive to the nature of the media used. The solvation study depicts the solubility of the optimized interacted molecule in different solvents (water, ethanol, methanol, heptane, cyclohexane, and CCl₄). Both the complexes glucosamine-uracil and glucosamine-thymine exhibited high dipole moment on polar solvent, and there is a steep decrease observed with the decreasing polarity of the medium. However, the dipole moment of glucosamine-thymine in different solvents was found to be greater than glucosamine-uracil and the electronic energy of glucosamine-uracil in different solvents was found to be greater than

Fig. 8 Comparison of electronic energy of glucosamine complex with thymine and uracil in **a** polar solvent and **b** non-polar solvent

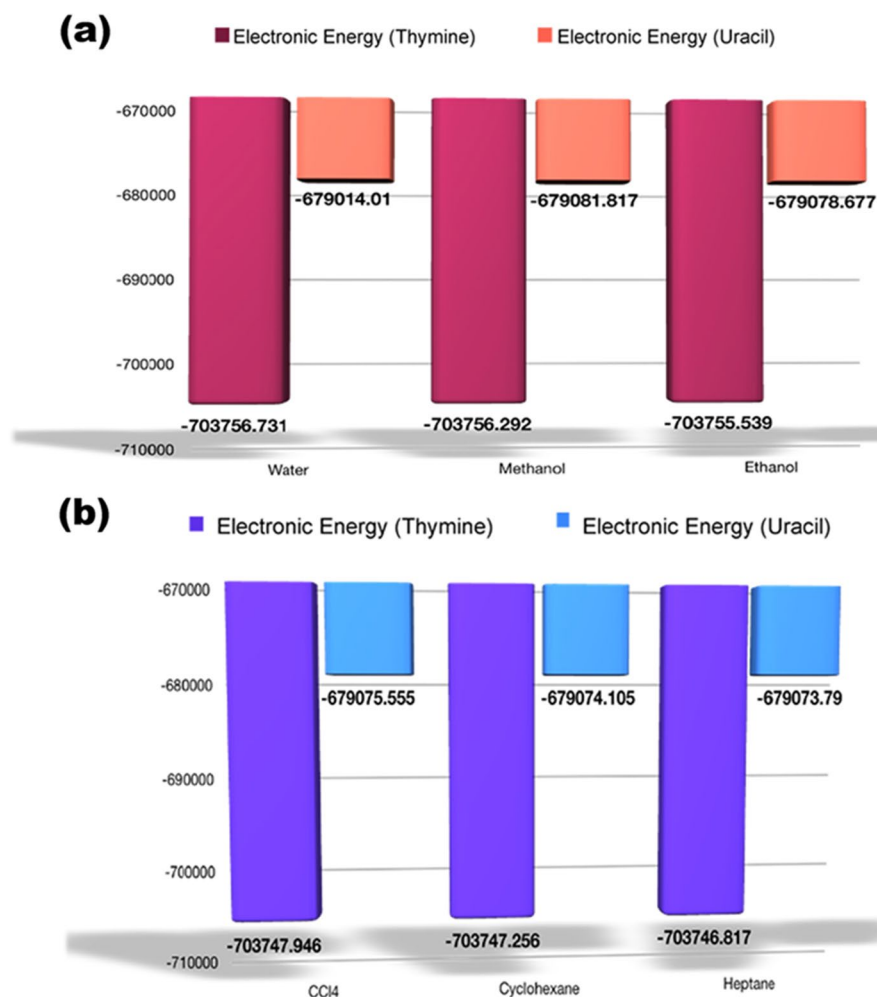


Table 7 Dipole moment in polar solvents

Solvents	Dipole moment (Thymine)	Dipole moment (Uracil)
Water	12.55	11.45
Methanol	12.48	9.85
Ethanol	11.28	9.21

Table 8 Dipole moment in non-polar solvents

Solvents	Dipole moment (Thymine)	Dipole moment (Uracil)
CCl ₄	11.16	8.30
Cyclohexane	11.06	6.78
Heptane	10.99	6.73

glucosamine-thymine. This study is believed to boost future research in designing of nucleic acid carriers based on glucosamine with new insights.

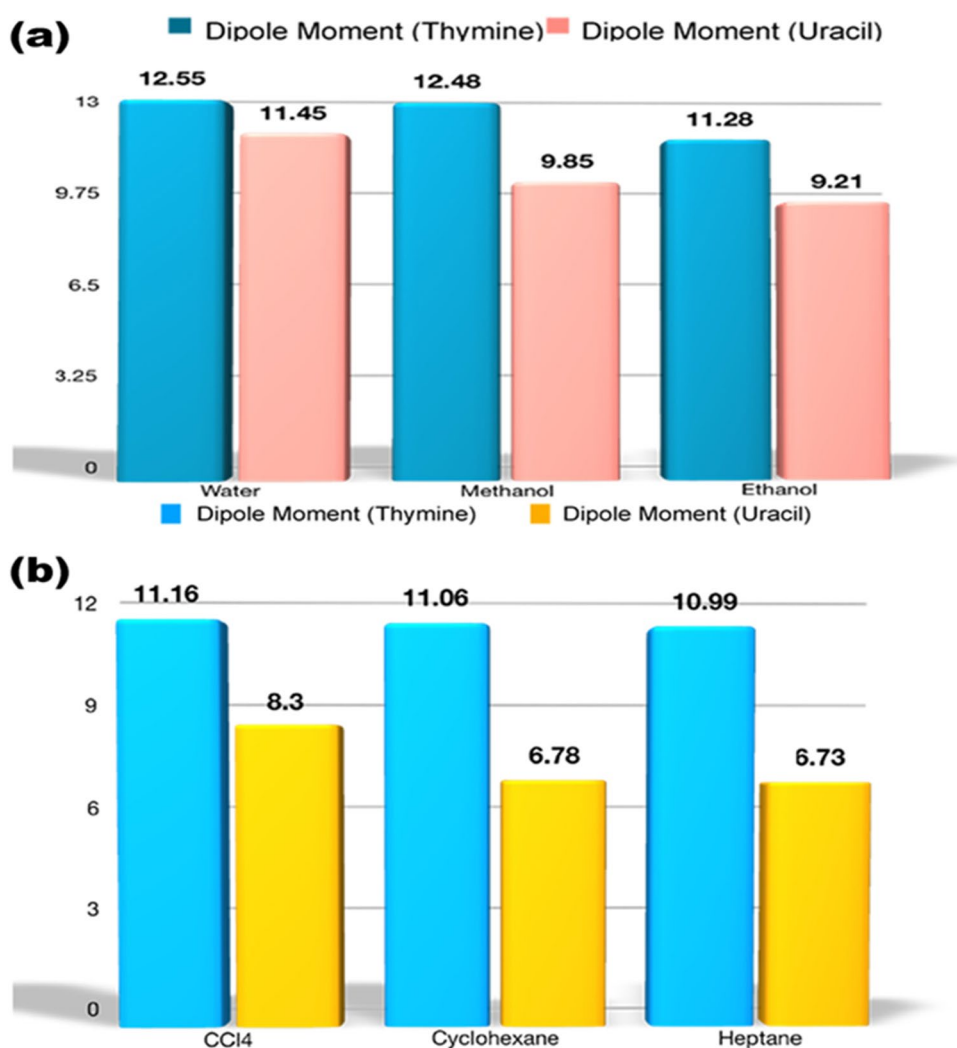
Acknowledgements I would like to thank the Madan Mohan Malaviya University of Technology, Gorakhpur, for providing us with the platform for doing M.Sc. Physics. I would like to acknowledge the help and support of Akanksha Singh, Research Scholar, Department of Physics, D.D.U. University, Gorakhpur.

Author contribution The outline of the research was done by all the authors. Aditi Srivastava did all the computation and wrote the first draft of the manuscript. B. K. Pandey provided resources and mentorship in research activity. Shivani Gupta reviewed and prepared the final draft of the manuscript. Saurav Mishra helped in plotting graphs and interpretation.

Data availability N/A.

Code availability N/A.

Fig. 9 Comparison of the dipole moment of glucosamine complex with thymine and uracil in **a** polar solvent and **b** non-polar solvent



Declarations

Competing interests The authors declare no competing interests.

Ethics approval N/A.

Consent to participate N/A.

Consent for publication N/A.

Conflict of interest The authors declare no competing interests.

References

- Jiang HL, Cui PF, Xie RL, Cho CS (2014) Chemical modification of chitosan for efficient gene therapy, 1st ed. Elsevier Inc
- Mao S, Sun W, Kissel T (2010) Chitosan-based formulations for delivery of DNA and siRNA. *Adv Drug Deliv Rev* 62:12–27. <https://doi.org/10.1016/j.addr.2009.08.004>
- Kumar S, Koh J, Kim H et al (2012) A new chitosan-thymine conjugate: synthesis, characterization and biological activity. *Int J Biol Macromol* 50:493–502. <https://doi.org/10.1016/j.ijbiomac.2012.01.015>
- Gardner J (1847) Original Papers. ON Ether-vapour, its medical and surgical uses. *Lancet* 49:431–434. [https://doi.org/10.1016/S0140-6736\(02\)87196-3](https://doi.org/10.1016/S0140-6736(02)87196-3)
- Golden M, Murrell B, Martin D et al (2020) Evolutionary Analyses of base-pairing interactions in DNA and RNA secondary structures. *Mol Biol Evol* 37:576–592. <https://doi.org/10.1093/molbev/msz243>
- Masoumi M, Jahanshahi M, Ahangari MG, Darzi GN (2020) Density functional theory study on the interaction of chitosan monomer with TiO₂, SiO₂ and carbon nanotubes. *Mater Chem Phys* 255:123576. <https://doi.org/10.1016/j.matchemphys.2020.123576>
- Creamer P (2000) Osteoarthritis pain and its treatment. *Curr Opin Rheumatol* 12:450–455. <https://doi.org/10.1097/00002281-200009000-00019>
- Taura LS, Ndikilar CE, Muhammad A (2017) Modeling the structures and electronic properties of uracil and thymine in gas phase and water. *Mod Appl Sci* 11:36. <https://doi.org/10.5539/mas.v11n12p36>
- Shishkin O, Gorb L, Chemistry JL-S (2009) undefined (2009) Conformational flexibility of pyrimidine rings of nucleic acid bases in polar environment: PCM study. *Springer* 20:743–749. <https://doi.org/10.1007/s11224-009-9477-1>
- Qi M, Zha G, Du DM et al (2010) Density functional theory study on the interaction between guanine-cytosine base pair and formamide. *Indian J Chem - Sect A Inorganic, Phys Theor Anal Chem* 49:1332–1338
- Farooqui H, Yadav A, Pandey BK (2021) Hydroxyl radical reactivity with cytosine and thymine: a computational study. *J Phys Conf Ser* 1849:1–5. <https://doi.org/10.1088/1742-6596/1849/1/012029>
- Samanta S, Raghunathan D, Mukherjee S, Sanyal B (2018) Dependence of HOMO-LUMO gap of DNA base pair steps on twist angle: a density functional approach. *J Biol Syst* 26:23–40. <https://doi.org/10.1142/S021833901850002X>
- Deka BC, Bhattacharyya PKR (2016) Reactivity of chitosan derivatives and their interaction with guanine: a computational study. *J Chem Sci* 128:589–598. <https://doi.org/10.1007/s12039-016-1064-6>
- Deka BC, Bhattacharyya PK (2017) DFT study on host-guest interaction in chitosan–amino acid complexes. *Comput Theor Chem* 1110:40–49. <https://doi.org/10.1016/j.comptc.2017.03.036>
- Comp T, Chem T, Yousefvand H et al (2017) Investigating chitosan – curcumin nanorings for containing fluorouracil department of medicinal chemistry, School of Pharmacy and Pharmaceutical Sciences, Isfahan Department of Pharmaceutics, School of Pharmacy and Pharmaceutical Sciences, Isfahan. *Turkish Comput Theor Chem* 1:6–12
- Fattahi A, Ghorat M, Pourjavadi A et al (2008) DFT/B3LYP study of thermochemistry of D-glucosamine, a representative polyfunctional bioorganic compound. *Sci Iran* 15:422–429
- Akour S (2017) Dynamic response of composite plate subjected to sudden heat flux. *Mod Appl Sci* 11:36. <https://doi.org/10.5539/mas.v11n12p36>
- Bakó I, Csókás D, Mayer I et al (2022) The influence of cations on the dipole moments of neighboring polar molecules. *Int J Quantum Chem* 122:1–12. <https://doi.org/10.1002/qua.26758>
- Tomasi J, Persico M (1994) Molecular interactions in solution: an overview of methods based on continuous distributions of the solvent. *Chem Rev* 94:2027–2094. <https://doi.org/10.1021/cr00031a013>
- Gupta S, Gupta AK, Pandey BK, Verma ML (2021) Correction to: Influence of polyethylene glycol plasticizer on the structural and electronic properties of PEO–NaI complex: a density functional study (*Journal of Molecular Modeling*, (2021), 27, 6, (155), DOI: 10.1007/s00894-021-04763-8). *J Mol Model* 27:1–11. <https://doi.org/10.1007/s00894-021-04969-w>

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